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SUPERSEDING
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FEDERAL SPECIFICATION

STEPLADDERS AND PLATFORM LADDERS, ALUMINUM MAGNESIUM, FIBERGLASS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers metal and fiberglass stepladders and platform ladders for general utility work.

1.2 Classification.

1.2.1 Types, classes, and kinds. Ladders furnished under this specification shall be of the following types, classes, and kinds as specified (see 6.1 and 6.2).

Type I - Industrial, heavy duty.
Type II - Commercial, medium duty.
Type III - Household, light duty.

Class A - Stepladders.
Class B - Platform ladders.

Kind 1 - Aluminum.
Kind 2 - Magnesium.
Kind 3 - Fiberglass.

1.2.2 Size. Ladders shall be of the sizes listed in table I, as specified (see 6.2).

TABLE I. Size

Type	Class	Kind	Size, folded length, feet											
I	A	1	3	4	5	6	7	8	10	12	14	16	18	20
		2	3	4	5	6	7	8	10	12	14	16	18	20
		3	3	4	5	6	7	8	10	12	14	16	18	20
II	A	1	3	4	5	6	7	8	10	12				
		2	3	4	5	6	7	8	10	12				
		3	3	4	5	6	7	8	10	12				
III	A	1	3	4	5	6								
		2	3	4	5	6								
		3	3	4	5	6								

Type	Class	Kind	Size, of ladder (feet)											
I	B	1	2	3	4	5	6	7	8	10	12	14	16	18	20
		2	2	3	4	5	6	7	8	10	12	14	16	18	20
		3	2	3	4	5	6	7	8	10					
II	B	1	2	3	4	5	6	7	8	10					
		2	2	3	4	5	6	7	8	10					
		3	2	3	4	5	6	7	8	10					
III	B	1	2	3	4										
		2	2	3	4										
		3	2	3	4										

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-S-781 - Steel Strapping, Flat.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-97 - Tape: Pressure-Sensitive, Adhesive, Filament Reinforced.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspections by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters Laboratories, Inc. (UL) Standards:

UL 184 - Standard for Portable Metal Ladders.

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 1285 Walt Whitman Road, Melville, LI, NY 11746.)

American National Standard Institute (ANSI) Standards:

A 14.2 - Safety Requirements for Portable Metal Ladders.
A 14.5 - Safety Requirements for Portable Reinforced Plastic Ladders.

(Applications for copies should be addressed to the American National Standard Institute, Inc., 1430 Broadway, New York, NY 10018.)

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 Each contractor for kind 1 and 2 ladders shall submit to the contracting agency proof that the ladder he proposes to supply under this specification conforms to the requirements of the Underwriters' Laboratories Inc., Standard UL 184 for Portable Metal Ladders. The label or listing mark of the UL may be accepted as evidence that the ladders conform to those requirements.

3.1.2 In lieu of the label or listing mark, the contractor for kind 1 and 2 ladders submit independent proof satisfactory to the contracting agency that the ladders conform to the requirements of UL 184 including methods of tests.

TABLE II. Weight of ladder

Ladder Size (feet)	Class A Max. Weight (lbs) Type I	Type II	Type III
3	15	11	--
4	21	13	10
5	26	15	12
6	32	18	14
7	36	24	--
8	42	30	--
10	52	36	--
12	62	42	--
14	75	--	--
16	88	--	--
18	98	--	--
20	108	--	--

3.2 Stepladder and platform ladder, general requirements.

3.2.1 Materials. Ladders shall be fabricated from any aluminum alloy of suitable temper for kind 1; any magnesium alloy of suitable temper for kind 2; or any fiberglass with characteristics to produce adequate compressive strength, stiffness, resistance to buckling for kind 3. All ladders must meet all the provisions of this specification. The fiberglass shall be in accordance with ANSI A 14.5 paragraph 4.

3.2.2 Use of steel. The use of steel for class A ladder hardware (i.e., spreaders, braces, etc.,) is acceptable provided the weight of the ladder does not exceed the values in table II.

3.2.3 Zinc. All steel surfaces shall be zinc coated.

3.2.4 Step spacing. The spacing of the steps shall be on 12 inch centers plus or minus 1/8 inch.

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jected to the examination and the tests shown in table I

3.2.5 Step and platform surfaces. Surfaces of steps and platforms designed for use in ascending and descending shall be corrugated, serrated, knurled, dimpled, or coated with skid-resistant material.

3.2.6 Burrs, bolts, rivets, and welds. All workmanship shall be free from burrs in excess of 1/64 inch. Bolt and rivet holes are to be accurate and within tolerances of good commercial metalworking practices. Rivets are to be set properly and to be free from structural defects. Welds are to conform to commercially accepted practices and to be free of defects.

3.3 Stepladder, other requirements.

3.3.1 Stepladder size. Stepladders longer than 20 feet shall not be supplied. The overall length of the ladder shall be measured along the front edge of the front side rail, including top cap and foot, with a tolerance of plus or minus 1/2 inch. Heavy duty, (type I) ladders shall be 3 to 20 feet, medium duty, (type II) ladders shall be 3 to 12 feet, and light duty, (type III) ladders shall be 3 to 6 feet in length (see 1.2.2 and 6.2).

3.3.2 Slope. The stepladder shall be designed and constructed to give a minimum slope of 3-1/2 inches per foot of length of the front section, and a minimum slope of 2 inches per foot of length of the back section.

3.3.3 Width and spread. The minimum clear width between the side rails at the top step shall be 12 inches. From top to bottom, the spread between the side rails shall increase a minimum of 1 inch per foot of side rail length.

3.3.4 Steps. Steps shall be parallel and level within tolerance specified.

3.3.5 Step width. The minimum width of the step for heavy and medium duty ladders shall be 3 inches and for light duty ladder shall be 2-1/2 inches.

3.3.6 Bucket shelf. When specified (see 6.2) a bucket shelf shall be provided. The bucket shelf shall be so fastened that it can be folded up when the ladder is closed.

3.3.7 Back section. The back section shall be designed with either horizontal or diagonal bracing.

3.3.8 Feet. The bottoms of the four rails, shall be provided with slip-resistant material for safety of the user.

3.3.9 Spreaders. The spreaders or locking device shall be of the pinch-proof design of sufficient size and strength to securely hold the front and back sections in the open position and shall be a component part of each stepladder. The pinch-proof spreader shall not be more than 6-1/2 feet above the base and shall have all sharp points and edges covered or removed to protect the user.

3.4 Platform ladder, other requirements.

3.4.1 Size. The size of a platform ladder is designated by the overall length as determined by the length of the front edge of the front side rail from the top of the platform to the base of the ladder, including any foot or shoe, with a tolerance of plus or minus 1/2 inch. Heavy duty (type I) ladders shall be 3 to 20 feet, medium duty (type II) ladders shall be 3 to 12 feet in length, and light duty (type III) ladders shall be 3 to 6 feet in length (see 1.2.2 and 6.2).

3.4.2 Slope. Platform ladders shall be so constructed that when in the open position, the slope of the front section shall not be less than 3-1/2 inches and the slope of the back section shall be a minimum of 1 inch for each 12 inch length of side rail.

3.4.3 Width and spread. The minimum clear width between the side rails at the platform level shall be 14 inches. From top to bottom the spread between the side rails shall increase a minimum of 1 inch per foot of side rail length.

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3.4.4 Platform. The platform shall be at least 20 inches (vertically) below the top of the ladder, the area of the platform shall not be less than 200 square inches. The platform shall not extend more than 1-1/2 inches beyond the back rail.

3.4.5 Top rail. The back legs and side rails of a platform ladder shall extend at least 20 inches of the verticle height above the platform and shall be connected with a top member to form a three-sides rail. Equivalent construction may be provided.

3.4.6 Spreaders. Spreaders shall be of the pinch-proof design and shall be provided on all ladders except ladders under 8 feet in length where the platform hinging device is designed to lock the unit open. The spreaders shall have all sharp points or edges covered or removed to protect the user.

3.5 Strength and stability requirements for stepladder and platform ladders.

3.5.1 Compression test. There shall be no failure or permanent deformation when tested in accordance with 4.4.1.

3.5.2 Side rail bending. The side rails shall not show any signs of failure or any permanent deformation when tested in accordance with 4.4.2.

3.5.3 Step bending. The step shall not show any signs of failure or any permanent deformation in excess of 1/100th of the clear width of the step between the side rail, when tested in accordance with 4.4.3.

3.5.4 Step to side rail shear strength. There shall be no failure of the fasteners attaching the step to the side rail when tested in accordance with 4.4.4.

3.5.5 Bucket shelf. There shall be no permanent deformation of the bucket shelf, and the ladder shall not tip over when tested in accordance with 4.4.5.

3.5.6 Stability. Side rails shall not lift from the floor when tested in accordance with 4.4.6.

3.5.7 Side rail cantilever. The side rail section shall return to its original position with no permanent deformation after having been tested in accordance with 4.4.7.

3.6 Marking. Permanent marking, including manufacturer's name, rated load capacity, and ladder type, shall be embossed on a metal component or attached to either metal or reinforced plastic component of the ladder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Sampling.

4.2.1 Dimensional and visual examination. Sampling for these characteristics shall be in accordance with MIL-STD-105 inspection level S-4, acceptable quality level (AQL) 2.5 percent defective.

4.2.2 Sampling for tests. Ladders shall be sampled in accordance with MIL-STD-105 inspection level S-1. Failure of any sample to pass any of the tests shall be cause for rejection of the lot.

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4.3 Examination.

4.3.1 Dimensional visual examination. Samples selected in accordance with 4.2.1 shall be examined for dimensional and visual defects in accordance with the requirements of table III below:

TABLE III. Classification of defects end item

Defects
Guard rail less than 20 inches above platform for platform ladder.
Type, grade, class, and size not as specified.
Slope less than 3-1/2 inches per foot for front rails.
Slope for back legs less than 2 inches per foot for stepladder (unless otherwise specified), or less than 1 inch per foot for platform ladder.
Length not within tolerance.
Clear width between side rails not as specified.
Spacing of steps not within tolerance.
Width of steps less than 3 inches for industrial and commercial ladders and less than 2-1/2 inches for household ladders.
Step surface not slip-resistant.
Insulating, nonslip pads missing or not riveted on bottom of front rails and back rails.
Bucket shelf missing when specified.
Platform area less than 200 square inches or more than 400 square inches.
Height of spreader less than 1/3 the way down from the top and more than 6 feet and 6 inches above the base.
Rivets not corrosion resistant.
Sharp edges or burrs greater than 1/32 inch high present.
Cracks, undercuts or in-line porosity in welds.
Damage or defects affecting function or serviceability.
Damage or defects not affecting function or serviceability.
Steel surfaces not completely zinc covered or not evenly zinc coated.

4.3.2 Inspection of preparation for delivery. An inspection shall be made to determine whether packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table IV. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 6 defects per hundred units.

TABLE IV. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior)	Omitted; incorrect; illegible, or improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of container.
Contents	Gross weight exceeds requirements of the applicable box specification.

4.4 Tests. Samples selected in accordance with 4.2.2 shall be tested as specified in 4.4.1 through 4.4.7.

4.4.1 Compression test. The compression test shall be conducted as specified in ANSI A 14.2 paragraph 6.2.1 for kind 1 and 2, and as specified in ANSI A 14.5 paragraph 6.3.1 for kind 3 (see 3.5.1).

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4.4.2 Side rail bending test. The side rail bending test shall be conducted as specified in ANSI A 14.2 paragraph 6.2.2 for kind 1 and 2, and as specified in ANSI A 14.5 paragraph 6.3.2 for kind 3 (see 3.5.2).

4.4.3 Step bending test. The step bending test shall be conducted as specified in ANSI A 14.2 paragraph 6.2.3 for kind 1 and 2, and as specified in ANSI A 14.5 paragraph 6.3.3 for kind 3 (see 3.5.3).

4.4.4 Step to side rail shear strength test. The step to side rail shear strength test shall be conducted as specified in ANSI A 14.2 paragraph 6.2.4 for kind 1 and 2, and as specified in ANSI A 14.5 paragraph 6.3.4 for kind 3 (see 3.5.4).

4.4.5 Bucket shelf test. The bucket shelf test shall be conducted as specified in ANSI A 14.2 paragraph 6.2.5 for kind 1 and 2, and as specified in ANSI A 14.5 paragraph 6.3.5 for kind 3 (see 3.5.5).

4.4.6 Stability test. The front, side, and rear stability of the ladders shall be tested as specified in ANSI A14.2 paragraphs 6.2.6, 6.2.7, and 6.2.8 for metal ladders and as specified in ANSI A 14.5 paragraphs 6.3.6, 6.3.7, and 6.3.8 for fiberglass ladders (see 3.5.6).

4.4.7 Cantilever bending test. The front and rear rail cantilever bending shall be tested as specified in ANSI A 14.2 paragraphs 6.2.9 and 6.2.10 for metal ladders and as specified in ANSI A 14.5 paragraphs 6.3.9 and 6.3.10 for fiberglass ladders (see 3.5.7).

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each ladder shall be folded (collapsed) with the bucket shelf when applicable, secured to prevent its projection beyond the outside edges of the legs. The bucket shelf shall be secured in this position, to the back legs by taping, tying with strong twine, or soft wire. The back legs shall be secured to the front side rails with filament reinforced tape conforming to PPP-T-97, type II. The ladders shall than be packed in quantities not to exceed the weight limitation, in crates conforming to PPP-C-650, type II, style A. Waterproofing of contents shall be in accordance with the appendix to the crate specification.

5.1.2 Commercial. The ladders shall be preserved and packed in accordance with normal commercial practice. The complete pack shall be designed to protect the item against damage during shipment, handling, and storage, to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification.

5.2 Unitization. When shipments to Government depots are full car or truck-load, the packed ladders shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 lbs in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. Packages, shipping containers, and unitized loads, (when applicable) shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Packages, shipping containers, and unitized loads, (when applicable) shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use.

6.1.1 Type I. Heavy duty ladders are intended for industrial use such as in construction or industrial maintenance.

6.1.2 Type II. Medium duty ladders are intended for light industrial or commercial use such as in offices or where heavier loads are not commonly associated with the use of these ladders.

6.1.3 Type III. Light duty ladders are intended for household use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and kind (see 1.2.1).
(c) Size (see 1.2.2 and 3.3.1).
(d) Bucket shelf if required (see 3.3.6).
(e) Selection of applicable level of packing (see 5.1).
(f) Whether civil or military marking is required (see 5.3).

MILITARY INTEREST:

The DOD has waived coordination of this document until further notice.

CIVIL AGENCY COORDINATING ACTIVITIES:

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